CLOUD & DEVOPS

Azure Core Services Visual Guide

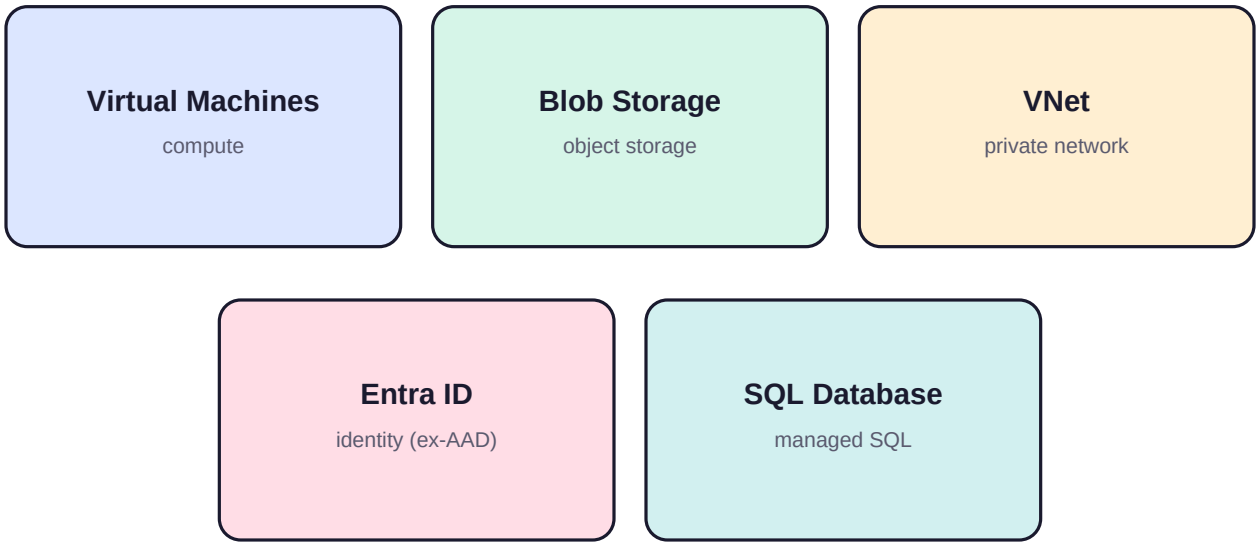
VMs, Blob Storage, VNet, Entra ID,
SQL Database — how they connect.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Meet the Five Core Services

Microsoft's equivalents of the cloud primitives.

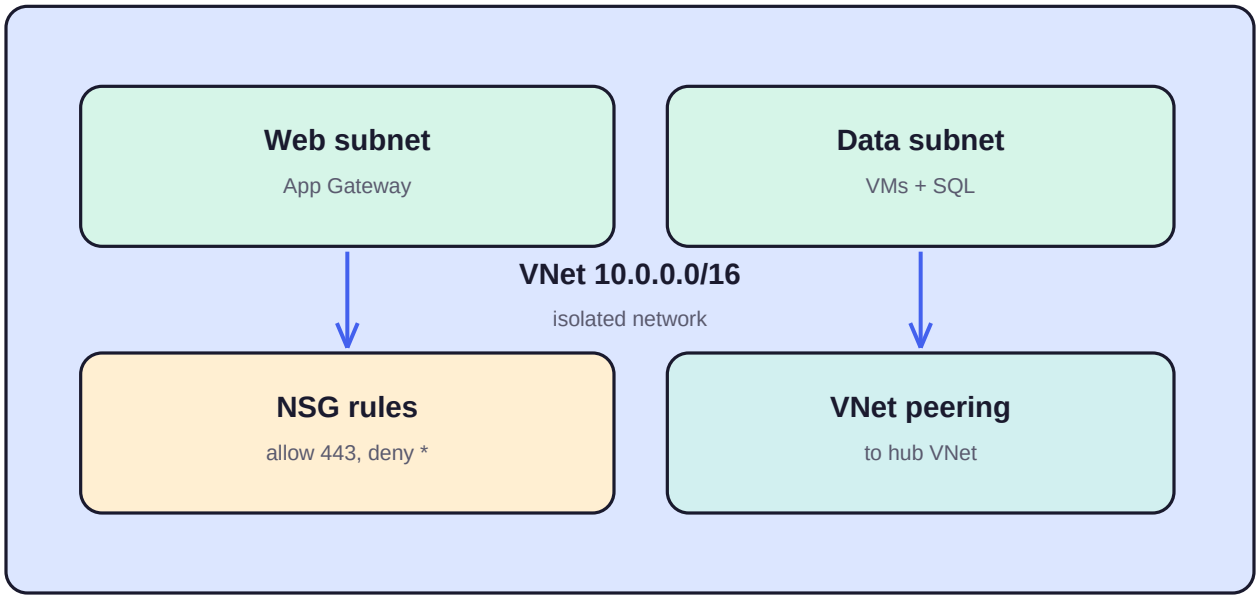


Everything in Azure lives in a Resource Group — the unit of billing, access, and lifecycle.

STEP 2/6

VNet: Subnets, NSGs, Peering

Isolated network; NSGs filter traffic per subnet or NIC.

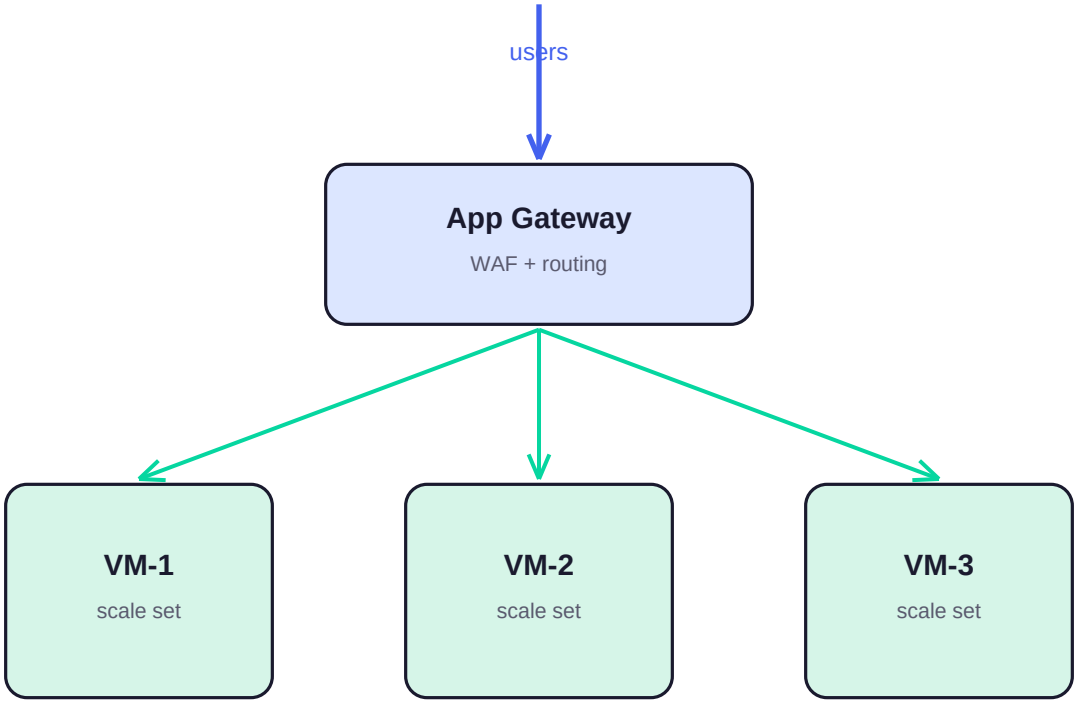


NSGs are stateful: allow inbound 443 and the return traffic flows automatically.

STEP 3/6

The Web Tier: VMSS + App Gateway

Virtual Machine Scale Sets behind a regional gateway.



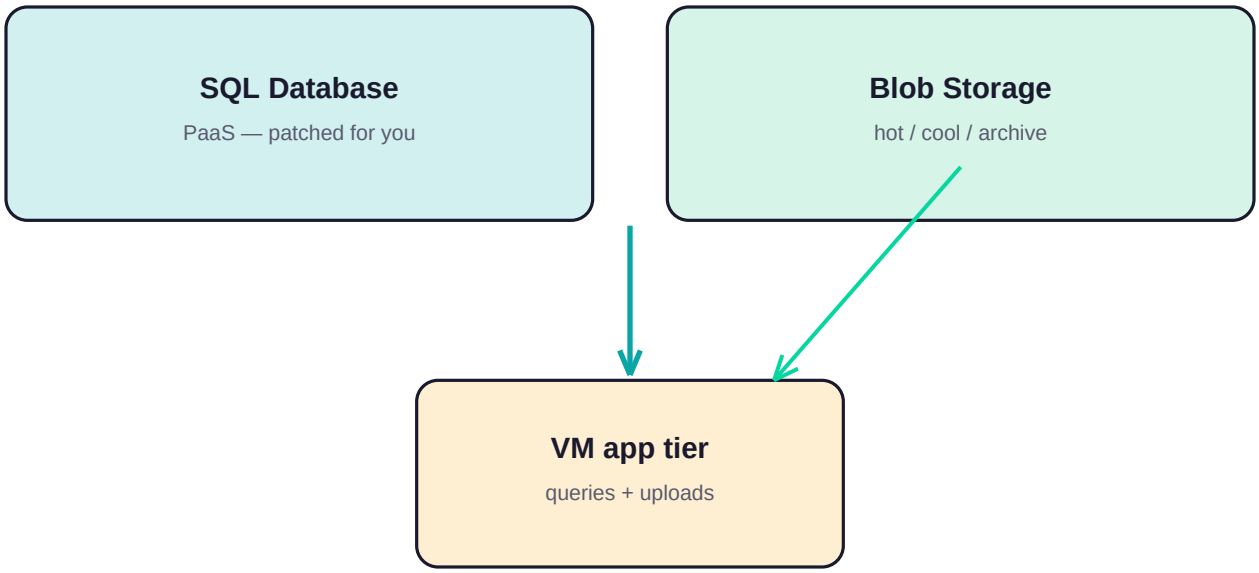
VM Scale Set: identical VMs from one image, autoscaled on metrics.

Application Gateway: layer-7 routing + Web Application Firewall.

STEP 4/6

Data: Blob + SQL Database

Hot/cool/archive tiers for blobs; managed SQL for rows.

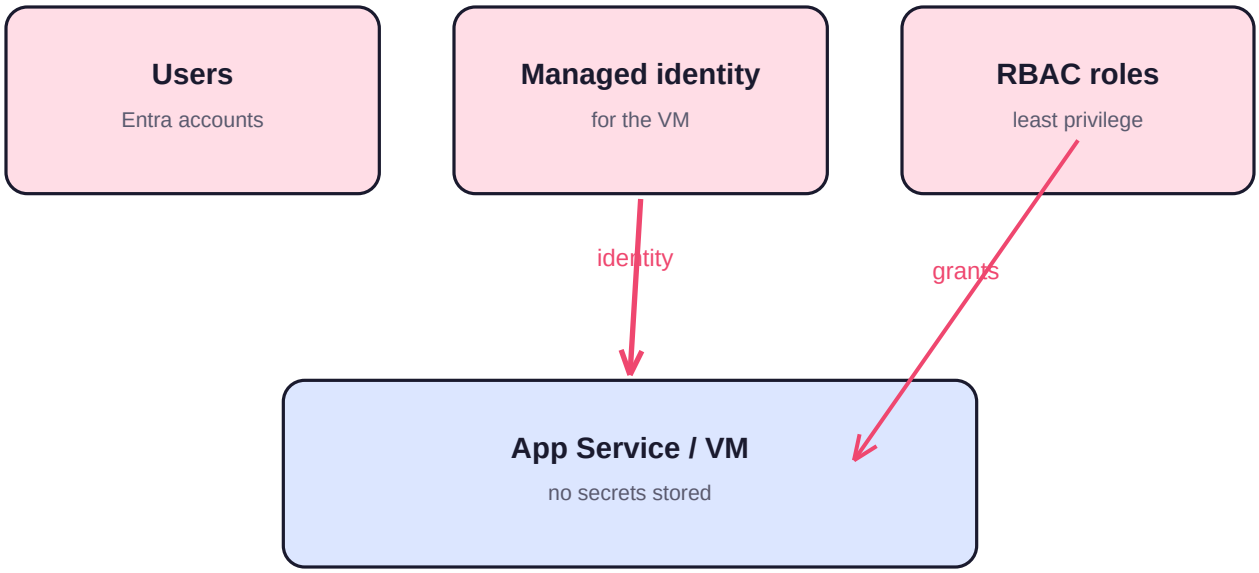


Blob lifecycle policies: auto-tier hot → cool → archive as data ages. Archive retrieval takes hours — plan restores.

STEP 5/6

Entra ID: Identity Without Passwords

Managed identities give Azure resources an Entra identity.

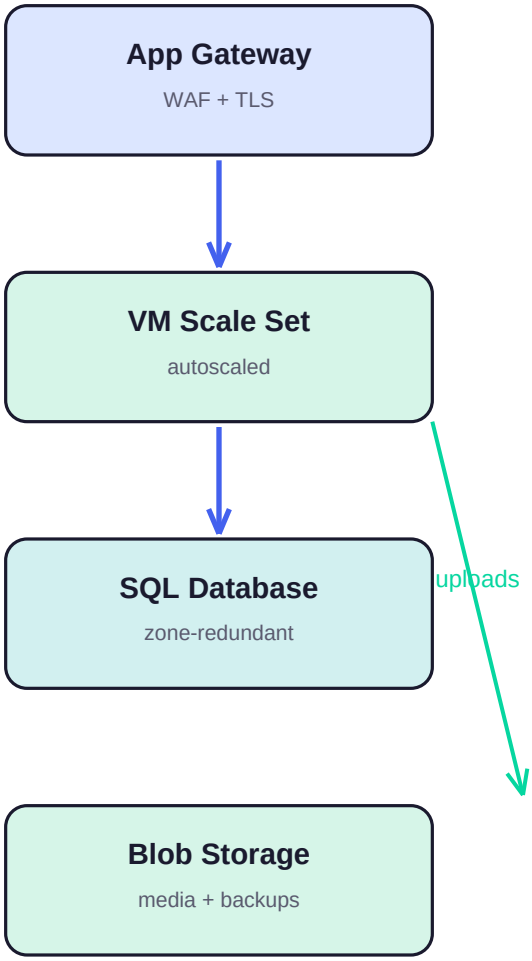


The app calls Azure services as itself — tokens are issued by the platform, rotated automatically.

STEP 6/6

Putting It Together: 3-Tier on Azure

The same architecture, Azure-flavored.



STEP ✓I✓

Key Takeaways

Azure Core Services — Remember

- Resource Groups are the unit of billing, access, and lifecycle — name them well.
- VM Scale Sets + Application Gateway = autoscaled web tier with WAF.
- SQL Database is PaaS (patched/managed); Blob tiers cut storage cost as data ages.
- Managed identities eliminate stored secrets — the platform issues tokens.
- NSGs + VNet peering give you a segmented, hub-and-spoke network.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev